

Homework 2

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Math 23

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1 2.3

Question 12. Newton's law of cooling states that the temperature of an object changes at a rate proportional to the difference between its temperature and that of its surroundings. Suppose that the temperature of a cup of coffee obeys Newton's law of cooling. If the coffee has a temperature of 200°F when freshly poured, and 1 min later has cooled to 190°F in a room at 70°F, determine when the coffee reaches a temperature of 150°F.

Solution. Newton's law of cooling gives the ODE

$$\frac{dT}{dt} = k(T - 70),$$

where $T(t)$ is the temperature of the coffee at time t and k is an arbitrary constant. Now, we solve the ODE as such

$$\frac{dT}{T - 70} = k dt \implies \ln |T - 70| = kt + C \implies T(t) = 70 + Ae^{kt}.$$

Applying the initial condition $T(0) = 200$,

$$200 = 70 + A \implies A = 130,$$

so

$$T(t) = 70 + 130e^{kt}.$$

Now let us find k with the condition $T(1) = 190$:

$$190 = 70 + 130e^k \implies e^k = \frac{120}{130} = \frac{12}{13} \implies k = \ln\left(\frac{12}{13}\right)$$

Finding t when $T = 150$:

$$150 = 70 + 130e^{kt} \implies e^{kt} = \frac{80}{130} = \frac{8}{13} \implies t = \frac{\ln(8/13)}{\ln(12/13)}$$

$$\approx \boxed{6.07 \text{ min.}}$$

Therefore, the coffee reaches 150°F after approximately 6.07 minutes. □

2 2.4

Question 3. Determine (without solving the problem) an interval in which the solution of the given initial value problem is certain to exist.

$$(4 - t^2)y' + 2ty = 3t^2, \quad y(-3) = 1$$

Solution. Let us first isolate y' so that we get the standard form as such:

$$y' + \frac{2t}{4 - t^2}y = \frac{3t^2}{4 - t^2}$$

Therefore,

$$p(t) = \frac{2t}{4 - t^2}, \quad g(t) = \frac{3t^2}{4 - t^2}.$$

Both $p(t)$ and $g(t)$ are not continuous when $t = \pm 2$. These points divide the t -axis into the three intervals $(-\infty, -2)$, $(-2, 2)$, $(2, \infty)$. Since the initial condition is $y(-3) = 1$ and $-3 \in (-\infty, -2)$, the theorem guarantees that a unique solution exists on

$$\boxed{-\infty < t < -2}.$$

□

Question 6. State where in the ty -plane the hypotheses of Theorem 2.4.2 are satisfied.

$$y' = \frac{\ln |ty|}{1 - t^2 + y^2}$$

Solution. Let

$$f(t, y) = \frac{\ln |ty|}{1 - t^2 + y^2}.$$

Then,

$$\frac{\partial f}{\partial y} = \frac{\frac{1 - t^2 + y^2}{y} - \ln |ty|(2y)}{(1 - t^2 + y^2)^2} = \frac{1 - t^2 + y^2 - \ln |ty|(2y)}{y(1 - t^2 + y^2)^2}.$$

In order for f and $\frac{\partial f}{\partial y}$ to be continuous, we need to consider certain scenarios. Firstly, $ty \neq 0$, so that $\ln |ty|$ is defined. Then,

$$t \neq 0 \quad \text{and} \quad y \neq 0.$$

Secondly, $1 - t^2 + y^2 \neq 0$, so that there is no division by zero in the denominator. This requires

$$y^2 \neq t^2 - 1.$$

Therefore, the hypotheses of Theorem 2.4.2 are satisfied in any region of the ty -plane where

$$\boxed{t \neq 0, \quad y \neq 0, \quad \text{and} \quad 1 - t^2 + y^2 \neq 0.}$$

□

Question 10. Solve the given initial value problem and determine how the interval in which the solution exists depends on the initial value y_0 .

$$y' = 2ty^2, \quad y(0) = y_0$$

Solution. We solve the equation by separation of variables as such:

$$\begin{aligned} \frac{dy}{y^2} = 2t \, dt &\implies \int \frac{dy}{y^2} = \int 2t \, dt \\ -\frac{1}{y} = t^2 + C &\implies y = -\frac{1}{t^2 + C}. \end{aligned}$$

When $y(0) = y_0$, then

$$y_0 = -\frac{1}{0^2 + C} \implies C = -\frac{1}{y_0}.$$

Therefore,

$$\boxed{y = -\frac{1}{t^2 - 1/y_0} = \frac{y_0}{1 - y_0 t^2}.}$$

If $y_0 > 0$, the denominator vanishes at $t = \pm 1/\sqrt{y_0}$, and since the initial condition is at $t = 0$, the solution exists only within the interval

$$-\frac{1}{\sqrt{y_0}} < t < \frac{1}{\sqrt{y_0}}.$$

If $y_0 < 0$, we have $-1/y_0 > 0$, so $t^2 - 1/y_0 = t^2 + |1/y_0| > 0$ for all t . The denominator never vanishes, and the solution exists on

$$-\infty < t < \infty.$$

Lastly, if $y_0 = 0$, then $y' = 0$ and $y = 0$ is the solution, which exists on

$$-\infty < t < \infty.$$

□

3 2.5

Question 5. Sometimes a constant equilibrium solution has the property that solutions lying on one side of the equilibrium solution tend to approach it, whereas solutions lying on the other side depart from it (see Figure 2.5.9). In this case the equilibrium solution is said to be **semistable**.

- (a) Consider the equation

$$dy/dt = k(1 - y)^2, \quad (19)$$

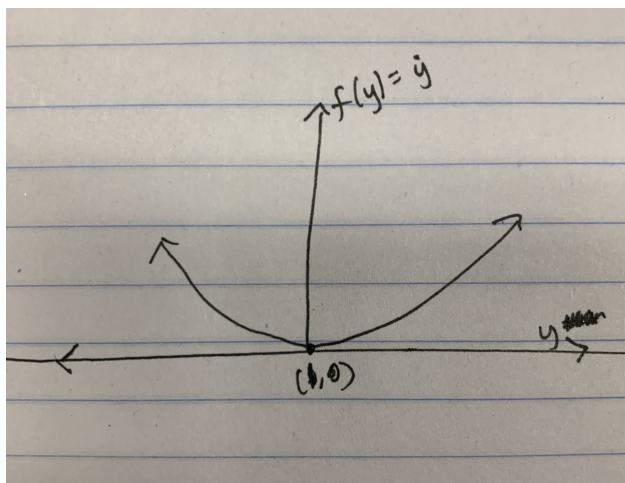
where k is a positive constant. Show that $y = 1$ is the only critical point, with the corresponding equilibrium solution $\phi(t) = 1$.

- (b) Sketch $f(y)$ versus y . Show that y is increasing as a function of t for $y < 1$ and also for $y > 1$. The phase line has upward-pointing arrows both below and above $y = 1$. Thus solutions below the equilibrium solution approach it, and those above it grow farther away. Therefore, $\phi(t) = 1$ is semistable.
- (c) Solve equation (19) subject to the initial condition $y(0) = y_0$ and confirm the conclusions reached in part (b).

Solution. For part (a), setting $dy/dt = 0$ gives

$$k(1 - y)^2 = 0 \implies (1 - y)^2 = 0 \implies y = 1.$$

Since $k > 0$, the only solution is $y = 1$, showing that $\phi(t) = 1$ is the only critical point and equilibrium solution. For part (b), I have attached the sketch below:



Function $f(y) = \frac{dy}{dt} > 0$ when $y \neq \phi(t) = 1$, hence proving that claims made in part (b).

For part (c), we solve by separation of variables:

$$\frac{dy}{(1-y)^2} = k dt$$

$$\implies \frac{1}{1-y} = kt + C.$$

Using the initial condition $y(0) = y_0$,

$$\frac{1}{1-y_0} = k(0) + C \implies C = \frac{1}{1-y_0}.$$

Therefore,

$$\frac{1}{1-y} = kt + \frac{1}{1-y_0} \implies y = 1 - \frac{1}{kt + \frac{1}{1-y_0}} \implies \boxed{y = 1 - \frac{1-y_0}{1+k(1-y_0)t}}.$$

As $t \rightarrow \infty$, if $y_0 < 1$ then $1 - y_0 > 0$ and $y \rightarrow 1$ from below, confirming solutions approach the equilibrium. If $y_0 > 1$ then $1 - y_0 < 0$ and the denominator vanishes, confirming solutions above $y = 1$ diverge away from the equilibrium. $\square$

Question 15. Suppose that a certain population obeys the logistic equation $dy/dt = ry(1 - (y/K))$.

- (a) If $y_0 = K/3$, find the time τ at which the initial population has doubled. Find the value of τ corresponding to $r = 0.025$ per year.
- (b) If $y_0/K = \alpha$, find the time T at which $y(T)/K = \beta$, where $0 < \alpha, \beta < 1$. Observe that $T \rightarrow \infty$ as $\alpha \rightarrow 0$ or as $\beta \rightarrow 1$. Find the value of T for $r = 0.025$ per year, $\alpha = 0.1$, and $\beta = 0.9$.

Solution. We can rewrite the logistic equation into its solved form as

$$y = \frac{y_0 K}{y_0 + (K - y_0)e^{-rt}}.$$

If we divide both sides by y_0 and substitute $y_0 = K/3$, we get

$$2 = \frac{K}{K/3 + (K - K/3)e^{-r\tau}}.$$

Cross-multiplying gives

$$\begin{aligned} \frac{2}{3} + \frac{4}{3}e^{-r\tau} &= 1 \\ \frac{4}{3}e^{-r\tau} &= \frac{1}{3} \implies e^{-r\tau} = \frac{1}{4}. \end{aligned}$$

Therefore,

$$\tau = \frac{\ln 4}{r} = \boxed{\frac{\ln 4}{0.025 \text{ year}^{-1}} \approx 55.45 \text{ years}}$$

For part (b), as $\alpha \rightarrow 0$ or $\beta \rightarrow 1$, the argument of $\ln$ tends to ∞ , so $T \rightarrow \infty$ as expected. Now, to find T at a set time, let us first divide both sides by K and then substitute $y_0 = \alpha K$:

$$\frac{y}{K} = \beta = \frac{\alpha K}{\alpha K + (K - \alpha K)e^{-rT}} = \frac{\alpha}{\alpha + (1 - \alpha)e^{-rT}}.$$

Since the values of α and β are given, we can now find T .

$$\begin{aligned} 0.9 &= \frac{0.1}{0.1 + (0.9)e^{-rT}} \\ 0.09 + 0.81e^{-rT} &= 0.1 \implies e^{-rT} = \frac{1}{81} \end{aligned}$$

Therefore,

$$T = -\frac{\ln |1/81|}{r} = \boxed{-\frac{\ln |1/81|}{0.025 \text{ year}^{-1}} \approx 176 \text{ years}}$$

□

4 2.6

Question 6. Determine whether the equation is exact. If it is exact, find the solution.

$$(ye^{xy} \cos(2x) - 2e^{xy} \sin(2x) + 2x) + (xe^{xy} \cos(2x) - 3)y' = 0$$

Solution. We can check if the equation is exact using Clairaut's Theorem. Set $M = ye^{xy} \cos(2x) - 2e^{xy} \sin(2x) + 2x$ and $N = xe^{xy} \cos(2x) - 3$. Then, let us find the partial derivatives:

$$\frac{\partial M}{\partial y} = \frac{\partial}{\partial y} [ye^{xy} \cos(2x) - 2e^{xy} \sin(2x) + 2x] = e^{xy} \cos(2x) + xye^{xy} \cos(2x) - 2xe^{xy} \sin(2x),$$

$$\frac{\partial N}{\partial x} = \frac{\partial}{\partial x} [xe^{xy} \cos(2x) - 3] = e^{xy} \cos(2x) + xye^{xy} \cos(2x) - 2xe^{xy} \sin(2x).$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, the equation is exact. Now, we seek $F(x, y)$ such that $F_x = M$ and $F_y = N$. Integrating N with respect to y :

$$F = \int (xe^{xy} \cos(2x) - 3) dy = e^{xy} \cos(2x) - 3y + h(x)$$

Now apply $F_x = M$:

$$F_x = ye^{xy} \cos(2x) - 2e^{xy} \sin(2x) + h'(x) = ye^{xy} \cos(2x) - 2e^{xy} \sin(2x) + 2x$$

$$h'(x) = 2x \implies h(x) = x^2.$$

Therefore, the general solution is

$$\boxed{e^{xy} \cos(2x) - 3y + x^2 = C.}$$

□

Question 10. Solve the given initial value problem and determine at least approximately where the solution is valid.

$$(9x^2 + y - 1) - (4y - x)y' = 0, \quad y(1) = 0$$

Solution. Set $M = 9x^2 + y - 1$ and $N = -(4y - x)$. The second partials $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} = 1$, so the equation is exact. We seek $F(x, y)$ such that $F_x = M$ and $F_y = N$. Integrating M with respect to x ,

$$F = \int (9x^2 + y - 1) dx = 3x^3 + xy - x + h(y)$$

$$F_y = x + h'(y) = -(4y - x) = x - 4y$$

Therefore,

$$h'(y) = -4y \implies h(y) = -2y^2,$$

and

$$F(x, y) = 3x^3 + xy - x - 2y^2.$$

and the general solution is:

$$3x^3 + xy - x - 2y^2 = C$$

Applying the initial condition $y(1) = 0$:

$$3(1)^3 + (1)(0) - (1) - 2(0)^2 = C \implies C = 2$$

So the solution is

$$3x^3 + xy - x - 2y^2 = 2$$

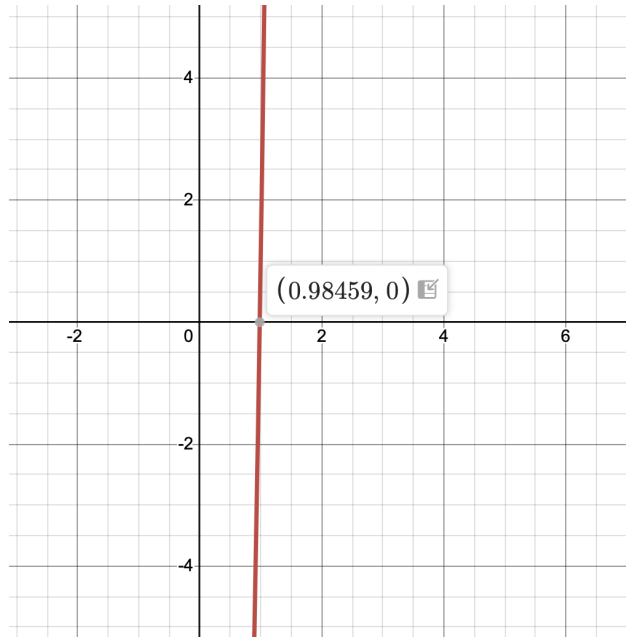
$$2y^2 - xy + \frac{x^2}{8} = 3x^3 - x - 2 + \frac{x^2}{8}$$

$$2\left(y - \frac{x}{4}\right)^2 = 3x^3 + \frac{x^2}{8} - x - 2$$

$$y = \pm \sqrt{\frac{3}{2}x^3 + \frac{x^2}{16} - \frac{x}{2} - 1} + \frac{x}{4} = \frac{x \pm \sqrt{24x^3 + x^2 - 8x - 16}}{4}.$$

However, since we define $y(1) = 0$, excluding the positive extraneous we get

$$\boxed{y = \frac{x - \sqrt{24x^3 + x^2 - 8x - 16}}{4}}.$$



The solution is not valid when $4x^3 + x^2 - 8x - 16 < 0$ because the square root of negative numbers is not defined in the real world. When plotting the polynomial, we see below that the only x -intercept is $x \approx 0.9846$. Therefore, the only solution valid is $x > 0.9846$.

□

Question 12. Find the value of b for which the given equation is exact, and then solve it using that value of b .

$$(ye^{2xy} + x) + bxe^{2xy} y' = 0$$

Solution. Set $M = ye^{2xy} + x$ and $N = bxe^{2xy}$. Let us use Clairaut's Theorem to solve b :

$$\frac{\partial M}{\partial y} = e^{2xy} + 2xye^{2xy}$$

$$\frac{\partial N}{\partial x} = be^{2xy} + 2bxye^{2xy}$$

Therefore,

$$e^{2xy} + 2xye^{2xy} = be^{2xy} + 2bxye^{2xy} \implies 1 + 2xy = b(1 + 2xy) \implies \boxed{b = 1.}$$

Hence, the equation is

$$(ye^{2xy} + x) + xe^{2xy} y' = 0.$$

Now, we seek to find the anti-derivative $F(x, y)$ such that $F_x = M$ and $F_y = N$. Integrating N with respect to y :

$$F = \int xe^{2xy} dy = \frac{1}{2}e^{2xy} + h(x)$$

Now apply $F_x = M$,

$$F_x = ye^{2xy} + h'(x) = ye^{2xy} + x.$$

Comparing both sides:

$$h'(x) = x \implies h(x) = \frac{x^2}{2}$$

Therefore the general solution is

$$\boxed{\frac{1}{2}e^{2xy} + \frac{x^2}{2} = C.}$$

□

Question 17. Show that if $(N_x - M_y)/M = Q$, where Q is a function of y only, then the differential equation

$$M + Ny' = 0$$

has an integrating factor of the form

$$\mu(y) = \exp \int Q(y) dy.$$

Solution. Recall that when utilizing integrating factors into exact first order ODE's, we can consider if μ is only a function of y so that

$$\mu_y M - \mu_x N = \mu(M_y - N_x) \implies \mu_y = \frac{N_x - M_y}{M} \mu.$$

But by substituting $Q = \frac{N_x - M_y}{M}$, we get that μQ is directly equal to μ_y . Hence, we evaluate this smaller differential equation.

$$\mu_y = \mu Q \implies \frac{d\mu}{\mu} = Q(y) dy$$

Integrating both sides gives

$$\ln |\mu| = \int Q(y) dy \implies \boxed{\mu = \exp \int Q(y) dy.}$$

□

Question 21. Find an integrating factor and solve the given equation.

$$y + (2xy - e^{-2y}) y' = 0$$

Solution. We are given a differential equation where we can define $M = y$ and $N = (2xy - e^{-2y})$. Let us see if there is an integrating factor that depends on y only such that

$$\mu_y M - \mu_x N = \mu(M_y - N_x) \implies \mu_y = \frac{N_x - M_y}{M} \mu.$$

Let us make substitutions given $M_y = 1$ and $N_x = 2y$ so that

$$\mu_y = \frac{2y - 1}{y} \mu = \left(2 - \frac{1}{y}\right) \mu \implies \frac{d\mu}{\mu} = \left(2 - \frac{1}{y}\right) dy.$$

Integrating both sides gives

$$\ln |\mu| = 2y - \ln |y| \implies \mu = \frac{e^{2y}}{y}.$$

Multiplying the original equation by $\mu = \frac{e^{2y}}{y}$ gives us the exact equation

$$e^{2y} + \left(2xe^{2y} - \frac{1}{y}\right) y' = 0.$$

Now set $M' = e^{2y}$ and $N' = 2xe^{2y} - \frac{1}{y}$. We seek $F(x, y)$ such that $F_x = M'$ and $F_y = N'$.

Integrating M' with respect to x give us

$$F = \int e^{2y} dx = xe^{2y} + h(y).$$

Now apply $F_y = N'$; so that

$$F_y = 2xe^{2y} + h'(y) = 2xe^{2y} - \frac{1}{y},$$

$$h'(y) = -\frac{1}{y} \implies h(y) = -\ln |y|.$$

Therefore the general solution is

$$\boxed{xe^{2y} - \ln |y| = C.}$$

□